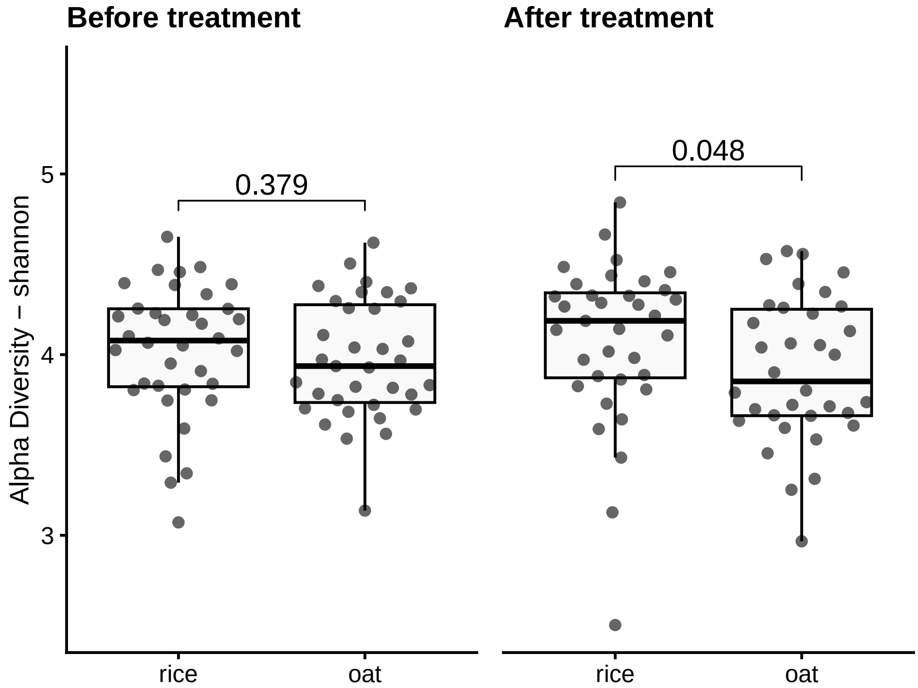
Alpha Diversity

Muluh

1 Group-wise comparisons

- **Index:** shannon
- **Group variable:** diet
- **Multiple testing correction:** FDR
- **Statistical test** Wilcoxon



1.1 Table of the top findings

1.1.1 Across-group

1.1.1.1 Before treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

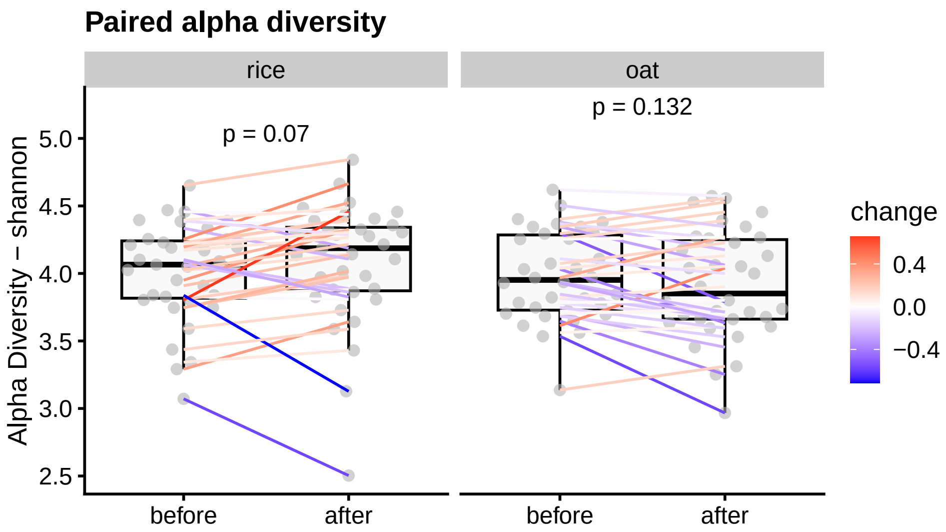
group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	4	4	0.02	0.379

1.1.1.2 After treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	3.9	4.1	0.06	0.048



1.1.2 Within-group

1.1.2.1 Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

diet	mean1	mean2	log2FC	p.value
rice	4	4.1	0.03	0.070
oat	4	3.9	-0.02	0.132

1.2 LMM

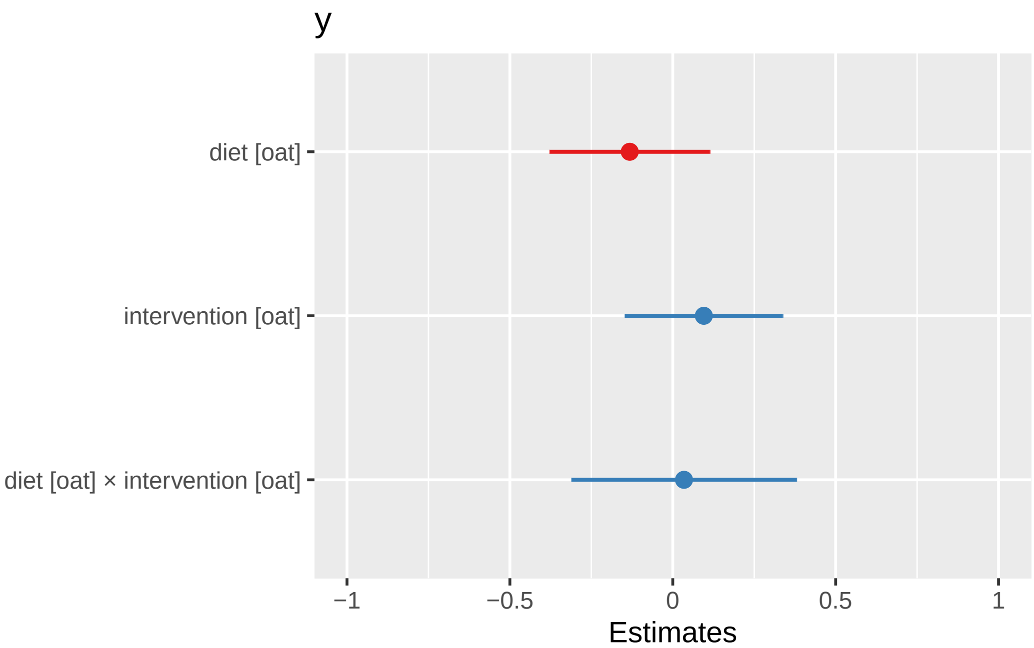
- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**
- Formula used: ~ **diet * intervention + (1 | id)**

Characteristic	before			after		
	Beta	95% CI[1]	p-value	Beta	95% CI[1]	p-value
(Intercept)	3.6	3.1, 4.0	<0.001	3.5	2.9, 4.1	<0.001
meal_group			0.3			0.3
1	—	—		—	—	
2	0.21	-0.03, 0.44	0.084	0.28	-0.02, 0.58	0.064
3	0.12	-0.12, 0.35	0.3	0.20	-0.09, 0.50	0.2
4	0.17	-0.06, 0.40	0.2	0.10	-0.19, 0.39	0.5
age	0.01	0.00, 0.01	0.2	0.01	0.00, 0.02	0.2

[1]CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI[1]	p-value
(Intercept)	3.5	-Inf, Inf	<0.001
meal_group			0.12
1	—	—	
2	0.23	-Inf, Inf	>0.9
3	0.16	-Inf, Inf	>0.9
4	0.13	-0.05, 0.32	0.2
age	0.01	0.00, 0.01	0.039

[1]CI = Confidence Interval



2 Alternate analyses

- **Beta (Estimate)**: Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI**: Range where the true beta is likely to fall, reflecting precision.
- **p-Value**: Tests if beta differs from zero; p<0.05p < 0.05p<0.05 suggests significance.

3 Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(1): relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AE2332 ... VK1336 VK2336
colData names(16): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(20): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```